scpPower M1 closure: plausibility note (v1.1.2)

13 October 2026. Amendments E/F close all five dataset/instrument rows. Earlier stops are preserved. No variance estimate was retuned to a source paper.

Specht. Raw PSM missingness 70.3%, mean 904.1 proteins/cell; quantification slope −0.042 [−0.045, −0.039]. The accepted ratio assay remains 79.5% missing with 624.5 proteins/cell and peptide−agreement cell CV 0.407. The paper's global estimate is approximately 79.5% missing and 622.7 proteins/cell; its 10.0% within−identified−peptide missingness has a different denominator. FLAG: the raw slope is weakly negative and does not support the usual abundance trend; it is retained exactly for the registered gate. The any−positive−PSM mask is earlier and more permissive than the paper's QC mask. The variance scale remains ratios; the predictor now measures raw abundance. Specht.

Leduc. Raw PSM missingness 63.7%, mean 1249.5 proteins/cell; slope 0.426 [0.423, 0.429]. The ratio input remains 70.8% missing; peptide−agreement cell CV 0.396 versus the paper's QC cutoff 0.41. The paper reports 997 proteins/cell after processing. FLAG: raw−mask completeness and coverage differ from the processed endpoint; raw protein ID groups and pre−QC PSMs do not reconstruct its final mask. Raw PSMs and ratio−based variance are explicitly separate inputs. Leduc.

Derks timsTOF / Q Exactive. Missing 33.0% / 59.6%; proteins/cell 1015.3 / 474.1; within−plex completeness 98.8% / 86.1%. timsTOF matches the published >98% within−plex completeness and approximately 1000 proteins/cell. FLAG: Q Exactive coverage is lower than that headline. Observed linear protein CV across cells 0.794 / 0.748; the paper's injection−repeat CV is a different quantity. Variance and curve estimates remain exactly as accepted, with file/label identity corrected. Derks.

Brunner. Missing 48.1%; proteins/cell 1284.2 (minimum 484 versus the paper's 600 cutoff, FLAG); median linear protein CV 0.350 versus reported 0.3, FLAG due to differing processing/QC. After per−cell median shifts, sigma2_e is total within−HeLa cell variance, including run noise. sigma2_b is zero with the reason "absorbed into sigma2_e: one cell per run". This is the identified label−free variance, not zero noise. Brunner.

Mapping and assumptions. Raw TMT PSMs are paired by exact IDs to the accepted protein/cell population, preserving cells and variance tiers without new QC thresholds. Walzik retains 23 healthy donors and 46 baseline observations. Overlap counts: 1862, 2579, 781, 603, 1553. Its proteomics is whole PBMC; immune subsets were flow cytometry, not sorted−subset proteomics. Donor composition differences are included; cell−line transfer remains an unvalidated proxy. k=1 is assumed, with sensitivity assumptions 0.75/0.5 and a two−proteome pilot standard recommended. Walzik; Cheung, Nature Methods.

Reproduction: inst/scripts/extract_tmt_psms.R, close_m1.R and report_final_m1.R; seed 20261013. Software: R=4.6.1; scp=1.22.0; limma=3.68.5; edgeR=4.10.5; scplainer=workflow in scp 1.22.0; no separate package